DAURES GREEN HYDROGEN VILLAGE

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4.4 TIMES Namibia (TINa) Energy System Model Results And Applications

OVERVIEW

This factsheet covers the results and applications of the TIMES Namibia (TINa) energy system model. The general TIMES energy system modelling frameworks and the development of TINa model are discussed in Factsheet 4.2 and 4.3 respectively; Namibia's energy profile overview, also crucial to understanding TINa model results is discussed in Factsheet 4.1.

The model explores the impacts of hydrogen and derivatives on Namibia's energy system, and how these play a role in achievement of Namibia's policy objectives. The amount of hydrogen and derivatives considered are based on the potentials of Daures green village project and other confirmed green hydrogen pilot projects in Namibia.

SCENARIO ANALYSIS

The following scenarios were analyzed with the TINa model:

Brief description	H2_Export	H2_Domestic	H2_Mix
Energy policy (achievement of 70% electrification rate,70% renewable energy share and 61% access to clean cooking by 2030; universal energy access by 2040)	√	√	√
Climate policy (energy related GHG emission reduction by 30% by 2030; achieve net zero emissions by 2050)	√	√	√
Hydrogen Strategy (export of hydrogen and derivatives)	√	x	x
Hydrogen Strategy with additional policies (export and domestic use of hydrogen and derivatives)	x	√	√

Table 1: TINa Scenario Matrix

Namibia 's green hydrogen strategy is export oriented, hence alternative scenarios H2_Domestic and H2_Mix explore the role hydrogen and derivatives can play in Namibia to support achievement of energy (Universal energy access and clean cooking) and environmental (30% energy related emission reduction by 2030 and net zero emission by 2050) goals.

H2_Domestic scenario targets the domestic use off all hydrogen and derivatives produced in different demand sectors in Namibia. similarly, H2_Mix target domestic use while exporting surplus.

FINAL ENERGY CONSUMPTION

With about 18 TWh of hydrogen, and 10 TWh of ammonia available as projected by confirmed pilot projects; In H2_Export scenario, the consumption of oil products persists throughout the model horizon, particularly in the transport, agriculture and industry sectors. In the alternative scenarios, however, the share of oil products gradually decreases as hydrogen, ammonia, and e-fuels emerge as substitute energy carriers.

The consumption of green ammonia, hydrogen, e-fuels, and hydrogen-based carriers across the scenarios is illustrated in the figures below. From the perspective of the objective function, the H2_Export scenario is the most cost-optimal, whereas the H2_Mix scenario demonstrates the highest level of energy efficiency.

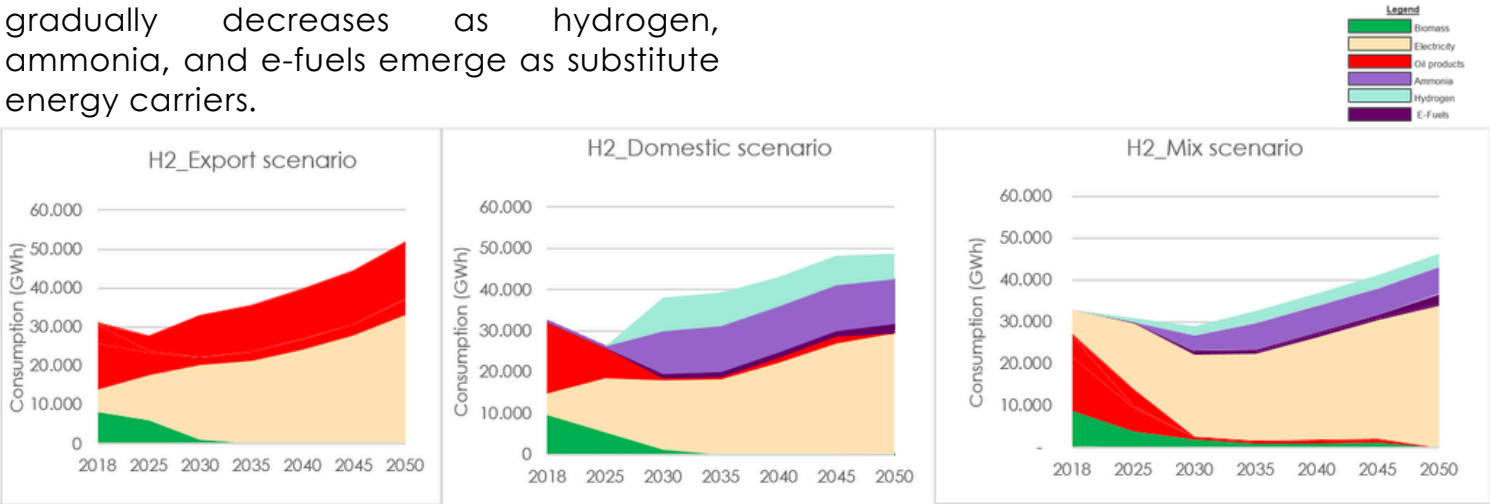

Figure 1. TINa model, final energy consumption by energy carrier by scenario

HYDROGEN AND DERIVATIVES CONSUMPTION

The consumption of hydrogen and derivatives by scenario is shown in figures below.

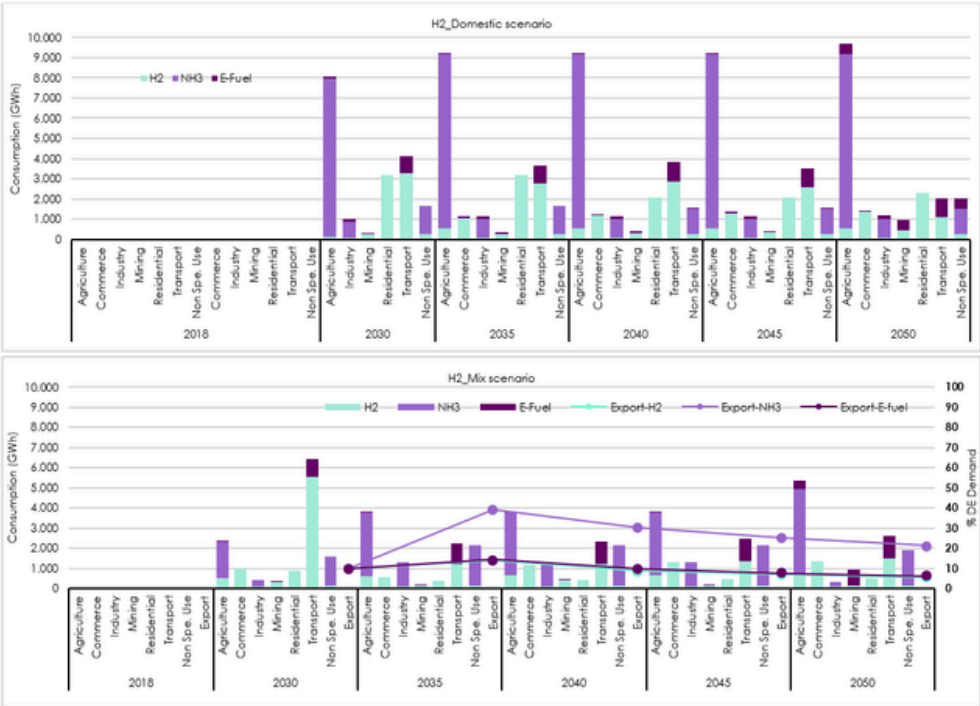

Figure 2. TINa model hydrogen and derivatives consumption by sector by scenario

Agriculture sector is projected to take up the most Ammonia, followed by industry, while direct consumption of hydrogen and e- fuels is most prominent in the transport sector. The available supply of hydrogen and derivatives can meet up to 10% of German (DE) demand of hydrogen and e-fuels, and about 40% of Ammonia by 2035.

The marginal price of Hydrogen of about 4.7 USD/kg and 0.5 USD/L for SAF by 2030 has been achieved by the model. The prices exclude storage and transport of energy carriers.

GHG EMISSIONS

Under the BAU scenario, achieving a 30% energy related emission reduction by 2030 is feasible if electricity consumption across sectors is increased. However, reaching zero emissions in the energy sector by 2050 may prove challenging due to the continued reliance on oil products, which account for more than 50% of final energy consumption in the base year, and the limited availability of alternative fuels to replace them. In contrast, these targets are attainable under the alternative scenarios.

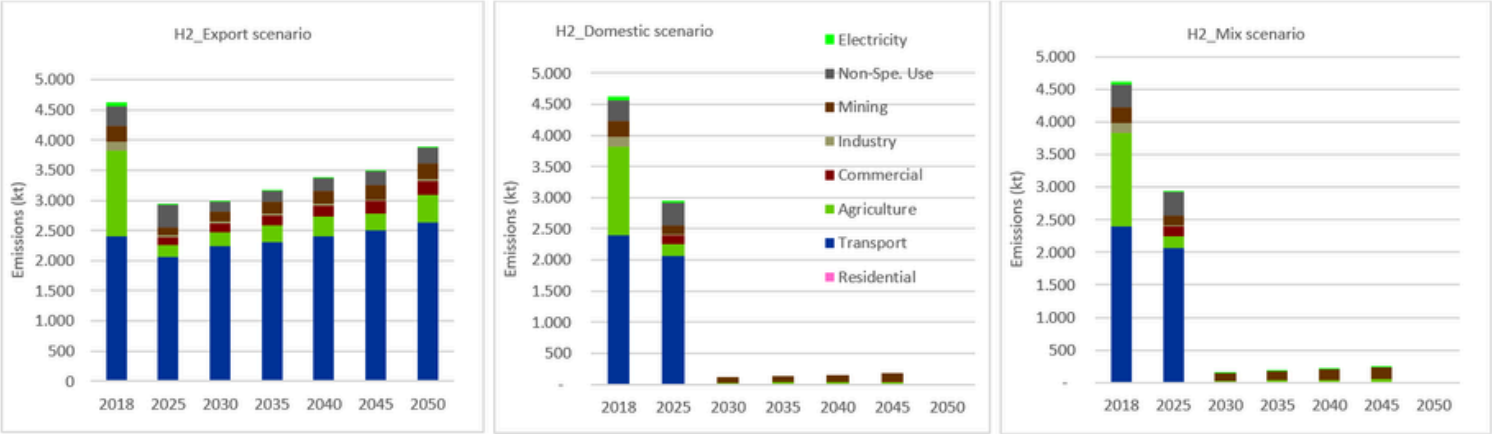

Figure 3. TINa model GHG emissions by sector by scenario

ROLE OF HYDROGEN AND DERIVATIVES IN NAMIBIA'S ENERGY SYSTEM

The model incorporates several pathways for hydrogen and derivatives use, including e-fuels and direct use of hydrogen in the transport sector, ammonia in the agriculture and industry sectors, hydrogen for electricity generation, and the direct use of hydrogen in the residential and commercial sectors.

CONCLUSIONS & SUMMARY

- TINa model explores the impacts of hydrogen and derivatives on Namibia's energy system.
- TINa can be applied to support decision-making and explore long-term options, not limited to hydrogen, energy and climate related targets.

Based on the analyzed scenarios, the following conclusions can be drawn:

- Hydrogen has the potential to contribute to improved energy access while supporting the decarbonization of multiple demand sectors.
- Namibia has the capacity to produce hydrogen and its derivatives for domestic consumption while simultaneously exporting surplus production.

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Further Fact Sheets covering topics from Project Descriptions, Use Cases, Techno-Economic analysis, Social Indicators, Energy System Analysis, Macro-Economic modelling, and more are available:

https://github.com/IER-Hy4Daures/Fact-Sheets